RIF On Chain
Stablecoin Protocol Collateralized with RIF and DOC

March 2026

Abstract — RIF On Chain (RoC) is a stablecoin protocol designed to issue USDRIF, a USD-pegged digital asset backed by on-chain collateral, without relying on custodial reserves or centralized issuers. To address the DeFi challenge of combining stability, capital efficiency, safe supply growth, and resilience under market stress, the protocol uses RIF and DOC in coordinated collateral buckets that support a single global USDRIF supply.

RIF absorbs volatility and turns it into growth potential through its residual balance token, RIFPRO. DOC contributes in two ways: its USD peg allows capital-efficient USDRIF issuance, and its residual balance token, ROCR, strengthens protocol-wide stability by absorbing liabilities from the RIF bucket under stressed conditions. The result is a censorship-resistant, transparent, over-collateralized stablecoin for the Rootstock ecosystem.

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1. Introduction

RIF On Chain (RoC) is a stablecoin protocol centered on USDRIF, a USD-pegged stablecoin backed by on-chain collateral. To support it, the system uses two collateral assets: RIF, the Rootstock Infrastructure Framework token, and DOC, Dollar on Chain.

USDRIF is backed by collateral whose total value exceeds the outstanding stablecoin supply. When the available collateral is insufficient to preserve the required safety margins, issuance is automatically restricted. The protocol operates entirely on-chain, without relying on custodial intermediaries or discretionary intervention to maintain the peg. Instead, stability is achieved through explicit collateral buffers and deterministic stabilization rules.

RoC organizes collateral into two coordinated economic substructures, referred to as buckets, which together support a single global USDRIF supply. Each bucket issues a residual token that represents the balance of that portion of the system. The RIF bucket issues RIFPRO, representing residual value backed by RIF collateral, while the DOC bucket issues ROCR, representing residual value backed by DOC collateral.

These two collateral classes serve complementary roles. RIF introduces growth potential and structured leverage through a volatile but economically productive asset. DOC contributes stability and scalable issuance capacity through a USD-denominated stablecoin collateral. Together, they allow USDRIF to remain over-collateralized while accommodating growing demand within the Rootstock ecosystem.

1.1 Rootstock and Bitcoin Alignment

Both collateral assets operate within the Rootstock ecosystem.

RIF is a utility token in the Rootstock ecosystem. DOC is a Bitcoin-backed stablecoin on Rootstock. By combining these assets, RoC remains fully on-chain and aligned with Bitcoin's security model through Rootstock's merged mining, inheriting security from Bitcoin's proof-of-work base instead of relying on an independent validator network.

USDRIF does not depend on custodial fiat reserves or centralized issuers. Its collateral is transparent and auditable on-chain.

This alignment preserves censorship resistance while enabling USD-denominated liquidity inside the Rootstock ecosystem.

2. System Architecture

RIF On Chain is implemented as a self contained on chain balance sheet organized around two collateral buckets, one for RIF and one for DOC. The protocol records the collateral held in each bucket, the amount of USDRIF attributed to it, and the Token Collateral issued against it. Although

USDRIF is a single fungible token across the system, liabilities are tracked separately at the bucket level.

Within this structure, USDRIF represents the senior claim on bucket value, while Token Collateral represents the residual claim. All protocol operations, including minting, redemption, and stabilization procedures, are designed to preserve the relationship between collateral value, USDRIF liabilities, and Token Collateral supply. When market conditions remain normal, the system operates through ordinary bucket operations. When a drop in collateral value causes a bucket's coverage to become stressed, stabilization procedures can reallocate collateral and liabilities across buckets.

The protocol architecture separates a small number of responsibilities across distinct components. Collateral accounting and token issuance are handled at the bucket level. System wide solvency is maintained through stabilization procedures coordinated by the Multi Collateral Guard, introduced in detail later. External market information, such as collateral prices, is introduced through oracle price feeds. Operational fairness and protection against adversarial execution are enforced by mechanisms such as the Operations Queue and the Flux Capacitor. Governance, in turn, controls protocol parameters but does not participate in the day to day execution of protocol operations.

2.1 High level system flow

Users interact with the protocol by choosing a bucket and depositing its corresponding collateral asset. In the RIF bucket, users may mint USDRIF, RIFPRO, or both. In the DOC bucket, users may mint USDRIF, ROCR, or both. These minting and redemption actions are the core operations of the protocol.

The two buckets do not operate under the same constraints. USDRIF can be redeemed from either bucket, allowing users to choose which collateral asset to receive, but only to the extent that the selected bucket has sufficient USDRIF liabilities available to process the redemption. In the RIF bucket, minting USDRIF may be limited if there is not enough over collateralization available, which depends on the residual balance provided through RIFPRO. Redeeming RIFPRO for RIF may also be limited if removing that collateral would push the bucket below its required collateralization threshold. In the DOC bucket, minting USDRIF is not subject to that same limitation, because DOC is designed to be worth one dollar and does not require over collateralization to support issuance, for that same reason, redeeming ROCR for DOC is not subject to any constraint. To help users operate around these limitations when needed, the protocol also allows combined operations, described in detail later.

When a drop in collateral value causes a bucket's coverage to become stressed, stabilization mechanisms can reallocate liabilities and collateral across buckets. These mechanisms are designed to preserve system integrity and support the continued normal operation of the protocol's core minting and redemption flows under adverse market conditions.

2.2 Tokens and Claims

RIF On Chain deals with three classes of tokens in each of its buckets.

Asset Collateral (AC): RIF and DOC

These are the exogenous assets, not issued by the protocol, used as economic backing.

Token Peg (TP): USDRIF

USDRIF is the USD-pegged stablecoin issued by the system.

Under normal operation, USDRIF can be redeemed for the equivalent value in collateral (minus protocol fees). Redeeming USDRIF burns the token and returns collateral (AC) from the corresponding bucket.

Token Collateral (TC): RIFPRO and ROCR

Each bucket issues its own Token Collateral (TC), **RIFPRO**, representing the residual balance of the RIF bucket and **ROCR**, representing the residual balance of the DOC bucket.

Token Collateral represents the remaining claim on bucket collateral after accounting for the USDRIF liabilities attributed to that bucket. When collateral value increases relative to liabilities, the value of TC rises. When collateral value decreases, TC absorbs losses before the stablecoin is affected.

Redeeming TC withdraws collateral from the bucket. Because this reduces the collateral backing the system, TC redemption is subject to coverage constraints: collateral cannot be withdrawn if doing so would reduce the bucket below its target collateralization ratio.

2.3 Buckets and Collateral Bags

RIF On Chain maintains two buckets, one for each collateral asset, the RIF bucket and the DOC bucket. Each bucket records its own collateral, the amount of USDRIF attributed to it, and the residual balance token issued against it, RIFPRO in the RIF bucket and ROCR in the DOC bucket. This structure allows USDRIF to remain a single fungible stablecoin across the protocol while preserving independent accounting for each bucket.

Bucket balance sheet

Each bucket can be understood as a self contained balance sheet. On one side is the collateral held in the bucket. On the other are the claims issued against it, namely the USDRIF attributed to that bucket and its residual balance token.

The total value of a bucket depends on how much collateral it holds and the market value of that collateral. That value must first support the USDRIF liabilities attributed to the bucket. Any remaining value belongs to the holders of the bucket’s residual balance token.

Coverage ratio

The solvency condition of a bucket is summarized by its coverage ratio, which measures the relationship between the value of its collateral and the amount of USDRIF attributed to it. Coverage is one of the protocol's main state variables, because it determines the admissible range of minting and redemption operations and when stabilization mechanisms must be activated.

When coverage is above one, a bucket contains enough collateral value to cover its attributed USDRIF liabilities. In RIF On Chain, however, the intended operating condition of the RIF bucket is not to remain merely above one, but significantly above it. Its coverage ratio is designed to remain several times higher than the minimum needed to cover outstanding USDRIF, so that the bucket maintains substantial over collateralization and a meaningful safety buffer against declines in the value of RIF collateral.

Because USDRIF is the senior claim on bucket value, that value must first satisfy stablecoin liabilities before any residual value can accrue to RIFPRO or ROCR holders. As coverage approaches one, the bucket becomes increasingly sensitive to changes in collateral value, which is why the protocol is designed to operate with a much larger cushion in the RIF bucket under normal conditions.

Residual claim and collateral token price

USDRIF represents the senior claim on the value stored in each bucket, while RIFPRO and ROCR represent the residual claims in their respective buckets. Their value therefore reflects the portion of bucket value that remains after the USDRIF liabilities attributed to that bucket have been accounted for.

This means the price of RIFPRO and ROCR is derived from the balance sheet of their respective buckets. After accounting for the value required to cover the bucket's USDRIF liabilities, the remaining value belongs to the holders of its residual token. As the value of the underlying collateral changes, the value of this residual claim changes proportionally more than the value of the stablecoin claim.

Because the RIF bucket is backed by a more volatile asset and operates with a different collateralization profile, the price of RIFPRO is likely to fluctuate much more than the price of ROCR. This difference follows from the structure of the two buckets and is discussed further in later sections. The formal pricing relationship is provided in Appendix A.

2.4 Multi-Collateral Guard

USDRIF is backed by the protocol as a whole, through both buckets and their collateral. For that reason, the buckets must remain linked, and the component that links their accounting is called the Multi Collateral Guard.

If the price of RIF or DOC falls and a bucket becomes stressed, the Multi Collateral Guard can shift part of the USDRIF backing from the degraded bucket to the healthier one. It does this by taking collateral from the stressed bucket, swapping it for the collateral of the healthy bucket, depositing that collateral into the healthy bucket, and transferring the corresponding USDRIF liability there as well.

This is how the protocol preserves system wide backing across both buckets rather than treating them as isolated pools. The detailed mechanics of this cross bucket transformation, impact for users, and the conditions under which it may occur, are described further in Section 5, Stabilization Mechanisms.

2.5 Protocol prices, and Price Oracles

The protocol distinguishes between different classes of prices according to their origin and role within the system.

Collateral token price (P_TC).

The price of Token Collateral is determined endogenously by the protocol from the bucket balance sheet. As shown previously, it corresponds to the residual claim on bucket value after stable token liabilities are accounted for.

This separation ensures that external market information is incorporated where necessary, through the collateral asset price, while the pricing of protocol-native tokens remains internally consistent and derived from the bucket state.

Collateral asset price (P_AC).

The market value of collateral is supplied to the protocol through the Money on Chain decentralized oracle infrastructure, known as OMOC.

The RIF bucket uses the RIF/USD price published by the RIF/USD OMOC job. This price is used to value RIF collateral and contributes to coverage calculations, issuance constraints, redemption constraints, and stabilization conditions.

The DOC bucket relies on DOC's USD-denominated value within the Money on Chain protocol. Because DOC's ability to track the U.S. dollar ultimately depends on the collateralization of the Money on Chain system, RoC also depends indirectly on an up-to-date BTC/USD price. That price allows the underlying Money on Chain protocol to evaluate the value of its Bitcoin collateral and maintain the conditions supporting DOC's USD peg.

OMOC operators therefore provide two price inputs relevant to RoC:

- RIF/USD, which RoC uses directly to value the collateral of the RIF bucket; and
- BTC/USD, which RoC depends on indirectly through the Money on Chain protocol and the DOC collateral used by the DOC bucket.

The operation and incentive model of the OMOC network, together with the jobs used by RoC, is described in Appendix C.

2.6 Decentralized Oracle Jobs and Protocol Execution

Certain protocol actions must be performed periodically or when predefined on-chain conditions are satisfied. RoC delegates the execution of these actions to the OMOC decentralized oracle network rather than relying on a dedicated set of protocol automators.

OMOC consists of independent node operators that coordinate off-chain to perform defined jobs. Depending on the job, the participating nodes reach consensus on data to be published on-chain or coordinate the execution of an on-chain function. Operators spend RBTC to submit transactions and receive rewards denominated in MOC tokens.

RoC uses the following OMOC jobs:

- **RIF/USD:** publishes the external RIF price used directly by the RIF bucket.
- **BTC/USD:** publishes the Bitcoin price used by the Money on Chain protocol and therefore indirectly supports RoC's use of DOC collateral.
- **TasksRunner:** executes periodic and condition-dependent protocol functions, including queued operations and stabilization procedures such as micro-liquidation and liquidation.

TasksRunner executors do not control protocol outcomes and have no discretion to modify the applicable rules. They submit transactions that invoke deterministic smart-contract functions. The contracts independently verify all execution conditions and reject calls that do not satisfy the protocol rules.

A more detailed description of OMOC participation, rounds, job allocation, transaction funding, and rewards is provided in Appendix C.

2.7 Protocol Operations

The protocol exposes several operations that modify bucket balances and token supplies.

Mint TP (Mint USDRIF)

A user deposits collateral into a bucket and receives newly issued USDRIF.

Redeem TP (Redeem USDRIF)

A user burns USDRIF and receives the equivalent value in collateral from the bucket.

Mint TC and TP

If minting USDRIF would require additional collateral to maintain the bucket's coverage ratio, the user may perform a Mint TC and TP operation. In this operation the user supplies the additional collateral required while simultaneously minting USDRIF.

Redeem TC and TP

If redeeming TC alone would reduce the bucket below its target coverage ratio, the user may perform Redeem TC and TP, redeeming both TC and USDRIF in the same operation. The USDRIF redemption reduces the collateral requirement, allowing the TC redemption to proceed.

Swap TC for TP and Swap TP for TC

The protocol also allows swaps between USDRIF and Token Collateral within each bucket: Swap TC for TP, Swap TP for TC.

These swaps depend on the amount of USDRIF attributed to the bucket where the swap occurs.

3. The RIF Bucket

The RIF bucket is the primary collateral engine of the protocol. It supports the issuance of USDRIF using RIF as collateral, and it issues the bucket's Token Collateral, RIFPRO, which represents the residual balance of the bucket.

RIF is a core infrastructure utility token within the Rootstock ecosystem. Its utility and integration suggest long term growth potential relative to USD denominated assets. RoC allows RIF to serve as collateral for the issuance of a USD-pegged stablecoin while maintaining over-collateralization.

When users deposit RIF into the bucket, they can mint USDRIF or mint RIFPRO. USDRIF represents the senior claim of the bucket, while RIFPRO represents the residual claim after accounting for the USDRIF liabilities attributed to the bucket.

Because RIF is volatile and its market capitalization remains relatively small compared to major cryptoassets, the RIF bucket operates with significant over collateralization buffers. These buffers protect USDRIF by ensuring that price declines in RIF are absorbed first by the residual balance of the bucket, rather than by the stablecoin.

RIFPRO behaves as a technical volatility absorber for fluctuations in the RIF collateral backing the bucket. When the price of RIF increases, the value of the bucket rises while the USDRIF liabilities attributed to it remain unchanged, so the residual value attributed to RIFPRO increases proportionally more. When RIF declines, losses are reflected in RIFPRO before they threaten USDRIF solvency. The formal relationship between coverage and Token Collateral leverage is provided in Appendix A.

RIF volatility and market depth impose practical limits on how much USDRIF can be safely issued using RIF collateral alone. This is why the protocol also supports issuance backed by DOC collateral, which provides additional capacity while preserving the system's solvency and stability.

The RIF bucket therefore performs two functions simultaneously. It provides collateral backing for the issuance of the stablecoin, and it allows RIF holders to take leveraged exposure to the performance of the RIF asset through the RIFPRO token.

3.1 Issuance and Redemption Dynamics

Operations involving the RIF bucket must maintain the target coverage ratio defined for the bucket. When a user deposits RIF and requests the issuance of USDRIF, the protocol verifies that the resulting collateralization remains above the required threshold.

If sufficient excess collateral exists in the bucket, USDRIF can be minted directly. If the operation would reduce coverage below the target level, the user may perform a Mint TC and TP operation instead. In this operation the user provides additional RIF collateral while minting USDRIF, increasing the residual balance of the bucket through the simultaneous issuance of RIFPRO.

The same logic applies in the opposite direction. Redeeming RIFPRO alone may not always be possible if doing so would reduce coverage below the target ratio. In those situations, the user may perform a Redeem TC and TP operation, redeeming both RIFPRO and USDRIF in the same transaction. In practice, this means the user is expected to obtain the required USDRIF from a third party or in a secondary market, from someone already holding USDRIF and therefore holding part of the bucket's outstanding liabilities. Burning that USDRIF reduces the bucket's liabilities and allows the corresponding RIFPRO redemption to proceed without violating coverage requirements.

The bucket also allows swaps between USDRIF and RIFPRO through the operations Swap TC for TP and Swap TP for TC. These swaps depend on the amount of USDRIF currently attributed to the bucket and the resulting coverage conditions.

3.2 Incentives for collateral, and Fee Flows

In the RIF bucket, minting and redemption operations generate fees that contribute to the economic activity of the protocol.

The RIF bucket also applies a tcInterestRate, which represents a periodic cost applied to the Token Collateral of the bucket. This rate is collected from RIFPRO holders and is implemented by periodically deducting value from the collateral backing the RIF bucket. The collected value contributes to the economic flows of the protocol and helps support the broader incentive structure of the system. A portion of the minting and redemption fees gathered by the RIF bucket is reintroduced as RIF collateral, which has the effect of raising the RIFPRO price. Initially, 25% of these fees are reinjected, a percentage that is subject to change via a governance proposal.

The RIF bucket includes a successFee mechanism tied to the appreciation of RIF relative to the USD. This mechanism is set at 0 at the moment, thus disabling it, but it may be enabled in the future.

When the price of RIF increases, a portion of this appreciation is captured by the protocol according to the configured successFee parameter. This mechanism allows the system to capture value surpluses produced by the increase in collateral value.

The successFee applies only in the RIF bucket and reflects the expectation that RIF may appreciate relative to the USD over time. Because the DOC bucket uses a USD-denominated collateral asset, the same mechanism is not applied there.

3.3 Price and Economic Protections

RIF is a volatile asset traded on external markets, operations involving RIF collateral are protected by several mechanisms designed to discourage economic attacks.

Depth-Weighted Average Price (DWAP)

The protocol uses a depth-weighted average price for RIF derived from exchange order books rather than a single spot price. Because the price reflects available liquidity rather than isolated trades, manipulating the oracle price requires sustained market impact rather than short-lived transactions.

Operations Queue

The operations queue delays execution of mint and redeem requests by a number of mined blocks. This delay discourages price manipulation by forcing any distorted market price to persist long enough for the queued operation to execute.

The queue also reduces oracle costs by allowing price updates to be written on-chain only when required.

Flux Capacitor

The Flux Capacitor limits the amount of minting and redemption that can occur within a recent block window.

Minting USDRIF is economically similar to selling RIF for USD without slippage, while redeeming USDRIF is similar to buying RIF. Frequent large mint or redeem operations could therefore create persistent arbitrage opportunities against external markets.

The Flux Capacitor restricts this behavior using two limits applied over a block span. An Absolute Accumulator tracks the total recent mint and redeem volume and enforces the Absolute Max Transaction Allowed (AMTA) limit. A Differential Accumulator tracks back-and-forth mint and redeem activity and enforces the Max Operational Difference Allowed (MODA) limit.

Both limits decay over time across a configured block span (BS) using a linear decay model, so that older activity progressively loses weight and transaction capacity is restored.

Appendix B is dedicated to explaining the Flux Capacitor in depth.

3.4 Initial Parameters

The RIF bucket is initially configured with the following parameters:

Target coverage ratio (Ctarg): 7x

successFee: 0 (disabled, but may be enabled in the future) tcInterestRate: 0.75% a year compounding weekly (0.000143211085680495 a week)

Operation queue delay: 3 blocks

Flux Capacitor parameters:

Block span (BS): 3512 (~24 hs)

Absolute Max Transaction Allowed (AMTA): 2,000,000 RIF

Max Operational Difference Allowed (MODA): 600,000 RIF

4. The DOC Bucket

The DOC bucket provides an additional source of collateral for the issuance of USDRIF using a USD-denominated asset.

DOC (Dollar on Chain) is a stablecoin issued by the Money On Chain protocol and designed to track the value of the US dollar through over-collateralization in Bitcoin. As a native asset of the Rootstock ecosystem, DOC combines Bitcoin-backed security with transparent on-chain collateralization.

By accepting DOC as collateral, RIF On Chain can satisfy demand for USDRIF even when the available supply of RIF collateral is limited or when the volatility of RIF makes additional issuance impractical. The DOC bucket therefore complements the RIF bucket by providing a stable collateral base that allows the supply of USDRIF to expand without relying exclusively on RIF.

Users who deposit DOC into the bucket receive the Token Collateral ROCR, which represents the residual balance of the DOC bucket after accounting for the USDRIF liabilities attributed to it.

4.1 Issuance and Redemption Dynamics

Issuance and redemption operations in the DOC bucket are simpler than in the RIF bucket because the collateral asset is already USD-denominated.

Users may mint USDRIF by depositing DOC into the bucket, and USDRIF can be redeemed by returning the token and withdrawing DOC. Because the bucket operates with a coverage ratio of one, these operations do not normally encounter the coverage constraints present in the RIF bucket, users can mint as much USDRIF as the DOC supply allows.

Joint operations such as Mint TC and TP and Redeem TC and TP remain available but are rarely required, since the bucket's collateralization model does not impose the same restrictions on TC redemption.

The bucket also allows swaps between USDRIF and ROCR through the operations Swap TC for TP and Swap TP for TC, subject to the amount of USDRIF attributed to the bucket.

4.2 Incentives for collateral, and Fee Flows

Because both DOC and USDRIF track the value of the US dollar, DOC collateral does not generate upside through price appreciation and does not introduce leverage effects when additional USDRIF is issued.

The DOC bucket nevertheless plays a key role in the protocol by enabling USDRIF to be minted against a stable USD-denominated asset. In practice, it provides the main path for expanding USDRIF supply independently of RIF volatility and market depth.

To encourage users to deposit DOC and mint ROCR, 25% of the fees generated by minting and redemption operations in the DOC bucket are reinjected into the bucket, increasing the residual balance attributed to ROCR holders. The reminder is distributed to MOC token stakers and OMOC Oracles according to the protocol's fee distribution rules. MOC token is the protocol's governance token, covered in section "6. Governance".

ROCR holders may also benefit from stabilization events involving the RIF bucket. When liquidation occurs in the RIF bucket, penalties applied to the stressed bucket can increase the residual balance of the receiving bucket.

Unlike the RIF bucket, the DOC bucket does not apply a **tcInterestRate** and will likely never have a **successFee**, since DOC is designed to maintain parity with the USD and is not expected to appreciate relative to it.

4.3 Collateralization Model and Economic Protections

Given DOC's parity with the USD, this bucket operates under a different collateralization and economic protection model than the RIF bucket.

The bucket is configured with a target coverage ratio of 1, meaning that one DOC of collateral is sufficient to back one USDRIF of liabilities. Unlike the RIF bucket, the DOC bucket does not require an over-collateralization buffer to absorb volatility in the collateral asset.

This structure allows USDRIF to be minted directly against DOC deposits while maintaining parity between the stablecoin and its collateral value.

The economic threat is also reduced, hence this bucket's operations queue delay is minimized to a single block, and the Flux Capacitor is disabled.

4.4 Initial Parameters

Target coverage ratio (Ctarg): 1

Operation queue delay: 1 block

Flux Capacitor: not applied

tcInterestRate: not applied

5. Stabilization Mechanisms

RIF On Chain maintains the solvency of USDRIF through conservative collateralization rules applied within each bucket. Under normal market conditions these rules ensure that fluctuations in collateral value are absorbed by the residual balance of the buckets without requiring corrective action.

The stabilization mechanisms described in this section exist as safeguards for extreme scenarios rather than as part of the expected day-to-day operation of the protocol. Multiple layers of protection are designed to prevent buckets from approaching the conditions that would trigger stabilization. These protections include conservative collateralization targets, depth-weighted average pricing for the RIF oracle, the operations queue, the Flux Capacitor, and protocol fees that discourage opportunistic arbitrage.

When market conditions deteriorate beyond the capacity of these protections, the protocol activates stabilization procedures. These procedures redistribute collateral and USDRIF liabilities between buckets in order to restore safe coverage levels and preserve the solvency of the stablecoin.

This process is coordinated by the Multi-Collateral Guard, which monitors bucket coverage ratios and triggers corrective actions when predefined thresholds are crossed. The guard operates through a sequence of mechanisms that begin with micro-liquidation and escalate to full liquidation if necessary.

Execution of these mechanisms is performed by external actors known as automators, which call the relevant smart contract functions when stabilization conditions are met.

5.1 Coverage Thresholds

Each bucket maintains a **coverage ratio**, defined as the market value of its collateral relative to the USDRIF liabilities attributed to it. This value is compared to a target parameter (**Ctarg**) that defines the collateral buffer required for normal operation.

To avoid overreacting to short lived price increases, the protocol evaluates coverage using an exponential moving average of the collateral price rather than only its latest value. This prevents temporary upward spikes from immediately expanding minting capacity and then weakening coverage if the price quickly reverses. It also smooths downward spikes so that they don't get mistaken as under-collateralization and prevent minting USDRIF when a collateral price rebound is likely to happen. This target coverage ratio adjusted by the exponential moving average (**Ctargema**) is formally defined in Appendix A.

When coverage remains above this target, the bucket operates normally and issuance and redemption proceed according to the protocol rules. If coverage approaches critical levels, stabilization mechanisms may activate to restore safe collateralization.

Stabilization occurs in stages. The first stage is **micro-liquidation**, which gradually reallocates collateral and liabilities across buckets. If deterioration continues, the protocol may proceed to **full liquidation**, performing a larger reallocation to restore safe system conditions.

5.2 Micro-Liquidation and Liquidation

The protocol stabilizes degraded buckets through a collateral reallocation process that transfers liabilities and collateral from a stressed bucket to a healthier one.

In both **micro-liquidation** and **liquidation**, collateral from the degraded bucket is sold and converted into the collateral asset of the receiving bucket. The corresponding USDRIF liabilities are transferred to the receiving bucket together with the acquired collateral. All collateral exchanges performed during these operations are executed through Uniswap pools.

A penalty is applied to the stressed bucket during this process (7 percent in the initial parameters). As a result, the bucket undergoing stabilization transfers more value than strictly required to cover the liabilities being moved. The excess value increases the residual balance of the receiving bucket.

The economic operation is the same in both cases. The difference lies in the scale and objective of the process.

Micro-liquidation transfers smaller portions of collateral and liabilities in order to restore safe coverage in the stressed bucket. This allows the bucket to continue operating normally while adequate collateralization is being re-established.

Liquidation is triggered under more severe deterioration. In this case the protocol attempts to remove the remaining collateral and liabilities from the degraded bucket as quickly as possible. Once liquidation completes, the bucket is considered permanently unusable.

Through this mechanism, the system makes sure that losses resulting from adverse market movements are absorbed by the residual balance of the stressed bucket, while attempting to preserve the solvency of USDRIF.

5.3 Possible scenarios

The stabilization mechanisms described above are designed to handle extreme market conditions.

The most plausible scenario involves a sharp decline in the price of RIF that reduces the coverage ratio of the RIF bucket. Although this scenario is unlikely, since the protocol's target coverage ratio for the RIF bucket was determined using historical RIF price data and configured with a buffer beyond the worst observed market declines.

In theory, stabilization could also occur if the DOC bucket were to lose its USD parity. While the protocol supports this scenario, it is considered highly unlikely. DOC has maintained its peg since its launch and is itself backed by an over-collateralized Bitcoin-based system.

For these reasons, micro-liquidation and liquidation should be understood as last-resort mechanisms that protect the solvency of USDRIF under extreme conditions.

6. Governance and Tokenomics

RIF On Chain operates within the governance framework of the Money on Chain (MoC) ecosystem. Protocol parameters, fee policies, and contract upgrades are governed through MoC governance using the MOC token.

The RIF On Chain protocol itself does not issue a governance token. Instead, governance authority is delegated to the existing MoC governance system.

Through governance processes defined by the Money on Chain ecosystem, MOC token holders may vote on decisions affecting the protocol, including:

- modification of protocol parameters such as coverage targets and stabilization thresholds
- adjustment of fee policies
- updates to operational limits and protection mechanisms
- approval of protocol upgrades
- addition of new buckets or pegged tokens.

This governance structure allows RoC to benefit from an established governance system while keeping the protocol architecture focused on stablecoin issuance and collateral management.

6.2 The MOC Token

The MOC token is the governance asset used across the Money on Chain ecosystem.

It is independent from the RIF On Chain protocol and is not required for ordinary interaction with the system. Users who mint or redeem USDRIF, RIFPRO, or ROCR do not need to hold MOC tokens.

However, MOC token plays an important role in coordinating governance decisions and aligning incentives across the ecosystem.

MOC tokens can be acquired independently in the market and may also be used to pay certain protocol fees.

6.3 Fee Flow and Revenue Distribution

RIF On Chain’s revenue distribution is designed to sustain the protocol and the services on which it depends. It rewards protocol supporters who stake MOC governance tokens, compensates OMOC nodes that keep prices updated and execute recurring background tasks, and funds Mimlabs’ stewardship work on software development, system monitoring, and security improvements. These allocations may be modified through the applicable governance process.

Minting and redemption fees are collected through both the RIF and DOC collateral buckets. When fees are paid in the bucket collateral asset, 25% is retained and reinjected into the corresponding bucket, strengthening its residual collateral value. The remaining fee flow is distributed as follows: 35% to Mimlabs; 17.5% to MOC stakers; 15% to RIF/USD price providers; 6.75% to BTC/USD price providers; and 0.75% to TasksRunner executors.

When fees are paid in MOC, the bucket allocation is converted back to the corresponding collateral asset before being returned to the bucket; other allocations may likewise be converted before delivery.

The RIF bucket also collects periodic interest from RIFPRO holders. This revenue is allocated 35% to Mimlabs, 35% to MOC stakers, 15% to RIF/USD price providers, 13.5% to BTC/USD price providers, and 1.5% to TasksRunner executors.

Rebalance fees from either bucket are converted to MOC and paid entirely to TasksRunner executors.

In addition to protocol revenue, RoC collects fees directly from users to fund the services required to process operations. Queue-execution fees are paid to the TasksRunner contract to compensate task executors. Price-update fees are split equally between the BTC/USD and RIF/USD OMOC oracle services, with each receiving 50%.

6.4 Governance Staking

Participation in governance requires staking MOC tokens.

Staking establishes an economic link between governance influence and long-term exposure to the system. Participants who stake MOC gain the ability to vote on governance decisions and may receive rewards generated by protocol activity.

Staked positions are subject to lockup periods defined by the governance framework. These constraints encourage long-term participation and reduce short-term opportunistic behavior in governance decisions.

6.5 Incentive Alignment

The governance and staking structure provides an incentive-alignment layer for the protocol.

Actors who participate in governance are economically exposed to the outcomes of the decisions they influence. By tying governance participation to staked tokens and distributing part of protocol revenue to governance participants, the system encourages decision-making that supports the long-term stability and growth of the protocol.

At the same time, governance and tokenomics remain separated from the core monetary mechanics of the protocol. The issuance, redemption, and stabilization behavior of the system operate independently from governance incentives, preserving the predictability of the stablecoin system.

7. Conclusion

RIF On Chain introduces a stablecoin architecture designed to combine resilience, transparency, and flexibility. By issuing a USD-pegged stablecoin backed by both a volatile asset (RIF) and a USD-denominated stablecoin (DOC), the protocol balances the advantages of over-collateralized systems with the scalability provided by stable collateral.

The system is structured around a bucket-based accounting model in which residual balance tokens absorb volatility while protecting the solvency of the stablecoin. Multiple layers of protection—including conservative collateralization targets, depth-weighted average pricing, transaction queues, and activity throttling mechanisms—are designed to prevent destabilizing behavior and discourage economic attacks.

In extreme market conditions, the protocol can redistribute collateralization burden between buckets through automated stabilization procedures. These mechanisms ensure that losses are absorbed by residual balance while preserving the integrity of the stablecoin.

Through this architecture, RIF On Chain provides a decentralized stablecoin that can expand with demand while maintaining strong safeguards against instability. The design allows the system to evolve through governance while preserving the core principles of transparent collateralization and predictable economic behavior.

Appendices

The following appendices provide a formal description of the protocol mechanics and protection mechanisms described throughout this document. These sections present the mathematical relationships and operational rules used by the system in a precise and unambiguous form.

While the main body of this document explains the design and reasoning of the protocol in prose, the appendices focus on formal definitions and explicit relationships between variables. The structure and notation are intentionally designed to be clear, systematic, and suitable for technical analysis, implementation reference, and automated interpretation by analytical or AI systems.

Appendix A — Formal Technical Specification

This appendix provides a formal description of the accounting structure and operational rules of the RIF On Chain protocol. The definitions and relationships described here correspond to the conceptual explanations presented throughout the whitepaper.

The protocol is modeled as a system that maintains a global stablecoin supply backed by multiple collateral buckets. Each bucket holds collateral assets and issues a corresponding Token Collateral (TC), which represents the residual balance of that bucket after accounting for its share of stablecoin liabilities.

A.1 Notation and State Variables

RIF On Chain maintains two collateral buckets: the RIF bucket and the DOC bucket. Each bucket is represented by a Collateral Bag (CB) that stores the collateral asset deposited in that bucket and tracks the Token Collateral issued against it. Stablecoin liabilities are tracked through Peg Containers (PC).

Token Peg. Notation: TP

Token Peg represents the stablecoin issued by the protocol.

In RIF On Chain, TP corresponds to USDRIF, the USD-pegged stablecoin issued by the system.

Token Collateral. Notation: TC

Token Collateral represents the collateral token issued by a Collateral Bag.

Each bucket issues its own Token Collateral: RIFPRO, issued by the RIF bucket, ROCR, issued by the DOC bucket

Asset Collateral. Notation: AC

Asset Collateral represents the underlying asset deposited in a Collateral Bag.

RIF On Chain accepts two collateral assets: RIF, used in the RIF bucket, DOC, used in the DOC bucket

Peg Container. Notation: PC

A Peg Container tracks the Token Peg liabilities associated with a specific Peg Container index.

Peg Containers record the amount of TP issued in the system.

In RIF On Chain, each bucket has exactly one Peg Container, corresponding to the USDRIF Token Peg.

Collateral Bag. Notation: CB

A Collateral Bag tracks the collateral state associated with a specific collateral asset.

A Collateral Bag records the amount of collateral deposited and the amount of Token Collateral issued by that bucket.

Amount of TP of Peg Container “i”. Notation: nTP(i)

nTP(i) represents the amount of Token Peg recorded in Peg Container “i”.

Amount of Collateral Asset in Collateral Bag. Notation: nACcb

nACcb represents the amount of Asset Collateral stored in a Collateral Bag.

Amount of Collateral Tokens in Collateral Bag. Notation: nTCcb

nTCcb represents the amount of Token Collateral issued by a Collateral Bag.

Protected state threshold. Notation: prot_thrld

prot_thrld represents the protected state threshold parameter used by the protocol.

AC price in TPi of the last operation. Notation: pACtp(i)lstop

pACtp(i)lstop represents the AC price expressed in TPi recorded during the last operation.

A.2 Peg Containers and Collateral Bags

RIF On Chain separates stablecoin liabilities from collateral accounting through two structures: Peg Containers (PC) and Collateral Bags (CB).

Peg Containers track Token Peg liabilities. Collateral Bags track the collateral assets deposited in the protocol and the Token Collateral issued against them.

Each bucket corresponds to a Collateral Bag. RIF On Chain maintains two Collateral Bags: The RIF Collateral Bag, which stores RIF and issues RIFPRO and the DOC Collateral Bag, which stores DOC and issues ROCR

Stablecoin liabilities are recorded in Peg Containers through the variable nTP(i). Collateral balances and Token Collateral supply are recorded in the Collateral Bags through the variables nACcb and nTCcb.

Operations such as stablecoin issuance and redemption update both the Peg Containers and the Collateral Bags in order to maintain the accounting relationship between stablecoin liabilities and collateral balances.

A.3 Stablecoin Supply Accounting

The protocol records Token Peg liabilities in Peg Container state variables. When Token Peg is issued, the protocol updates the Peg Container liability state, the Collateral Bag collateral state, and the "last operation" price state according to the following atomic process.

qTPi emission atomic process

TPiou[i] = TPiou[i] + otfPnLtpi; //to TPiiou is added on the fly P&L
nTPi = nTPi + qTPI; //nTPi is added to the amount that is issued
nCAcb = nCAcb + (qTPi / pACtpi); //to the collateral nACcb the value in CA of the qTPi that are issued is added
pACtplstop[i] = pACtpi; // the price of the last operation is updated

This process updates (i) the Peg Container liability amount (nTPi), (ii) the Collateral Bag collateral amount (nCAcb), (iii) the per-container "last operation price" (pACtplstop[i]), and (iv) the TP IOU tracking variable (TPiou[i]) through the on-the-fly P&L term (otfPnLtpi).

A.4 Token Collateral Accounting

Token Collateral (TC) represents the collateral token issued by a Collateral Bag. When Token Collateral is issued, the protocol updates both the amount of TC recorded for the bucket and the collateral amount stored in the Collateral Bag according to the following atomic process.

qTC emission atomic process

nTCcb = nTCcb + qTC; // add the newly issued Token Collateral (TC) to the total amount of TC collateral (nTCcb)
nCAcb = nCAcb + (qTC / pTCacreal); // add the value of the newly issued TC to the total amount of collateral (nCAcb) by dividing the TC amount by the real-time value of pTCacreal

This process increases the Token Collateral supply (nTCcb) by the quantity issued (qTC). The corresponding collateral value of the issued TC is added to the Collateral Bag by converting the TC amount using the real-time price variable pTCacreal.

A.5 Coverage Targets and Coverage Parameters

Coverage parameters determine how collateralization constraints are enforced by the protocol and how joint operations are evaluated.

Target coverage. Notation: Ctarg

Ctarg represents the target coverage parameter defined for the system.

Target coverage adjusted by exponential moving average. Notation: Ctargema

Ctargema represents the target coverage adjusted by an exponential moving average.

Coverage for joint operations. Notation: cobop

cobop represents the coverage level to which joint issuance and joint redemption operations are carried out.

Exponential Moving Average of Collateral Price. Notation: $P_{AC,EMA,t}$

The protocol computes an exponential moving average of the collateral asset price. This value is used when calculating the adjusted target coverage.

$$P_{AC,EMA,t} = \alpha \cdot P_{AC,t} + (1 - \alpha) \cdot P_{AC,EMA,t-1}$$

```
\begin{equation}
P_{AC,EMA,t} = \alpha \cdot P_{AC,t} + (1-\alpha)\cdot P_{AC,EMA,t-1}
\end{equation}
```

Where:

$P_{AC,t}$ is the spot price of the collateral asset at time t.

$P_{AC,EMA,t}$ is the exponential moving average of the collateral price.

EMA Smoothing Factor. Notation: α

The smoothing factor of the exponential moving average is defined as:

$$\alpha = \frac{2}{N+1}$$

```
\begin{equation}
\alpha = \frac{2}{N + 1}
\end{equation}
```

Where:

N is the effective EMA window length expressed in discrete time periods.

Larger values of N produce stronger smoothing, while smaller values make the EMA respond more quickly to price changes.

EMA-Adjusted Target Coverage. Notation: $CtargEMA$

The protocol defines an adjusted coverage target based on the ratio between the smoothed collateral price and the current spot price.

$$CtargEMA = Ctarg \cdot \frac{P_{CA,EMA}}{P_{CA}}$$

```
\begin{equation} C_{\mathrm{targEMA}} = C_{\mathrm{targ}} \cdot \frac{P_{AC,\mathrm{EMA}}}{P_{AC}} \end{equation}
```

This adjustment prevents short-term price fluctuations from relaxing the system's solvency constraints. The target coverage is technically allowed to temporarily fall below the nominal governance parameter after a sharp drop, to prevent underpricing collateral when a rebound is expected.

A.6 Bucket Value, Coverage and Protected State Conditions

Bucket Value

The value of a bucket corresponds to the market value of the collateral contained in its Collateral Bag.

Formally, the bucket value is defined as:

$$V_{bucket} = nACcb \cdot P_{AC}$$

```
\begin{equation}
V_{\text{bucket}} = TVL \cdot P_{AC}
\end{equation}
```

Where:

nACcb is the amount of AC (RIF,DOC) stored in the Collateral Bag.

P_{AC} is the price of the collateral asset expressed in USD.

This value represents the asset side of the bucket balance sheet.

Coverage Ratio

The coverage ratio measures the relationship between the value of the collateral backing a bucket and the Token Peg liabilities attributed to that bucket.

Formally:

$$C = \frac{TVL \cdot P_{CA}}{nTP}$$

```
\begin{equation}
C = \frac{TVL \cdot P_{AC}}{nTP}
\end{equation}
```

Where:

C is the coverage ratio.
nTP (USDRIF) liabilities in the bucket's Peg Container.

A coverage ratio greater than one indicates that the bucket contains sufficient collateral value to cover its Token Peg liabilities.

The protocol evaluates the collateralization state of each Collateral Bag in order to determine whether operations are allowed and whether the bucket enters a protected state. The protected state is defined by a threshold parameter that determines when the protocol must apply additional restrictions.

Protected state threshold. Notation: prot_thrld

prot_thrld represents the protected state threshold parameter used by the protocol to determine whether a Collateral Bag is in protected state.

When the coverage of a Collateral Bag falls below the protected state threshold (prot_thrld), the bucket enters a protected state. In this state, certain operations that would further reduce the collateralization of the bucket are restricted.

AC price in TPi of the last operation. Notation: pACtp(i)lstop

pACtp(i)lstop represents the price of the collateral asset expressed in TPi that was recorded during the last operation affecting Peg Container i.

This variable is updated during Token Peg emission operations and is used by the protocol when evaluating state transitions related to coverage and protected state conditions.

A.7 Joint Issuance and Joint Redemption

Joint operations adjust both Token Peg (TP) liabilities and Token Collateral (TC) balances in a Collateral Bag in a single operation. These operations are evaluated using the coverage parameter cobop.

About the proportionality of the Tokens

Being cobop the coverage to which the joint operation must be carried out, this is how we calculate the TP and TC required to maintain coverage. The relationship expressed in these formulas is the proportional adjustment between Token Collateral and Token Peg during joint operations so that the resulting state satisfies the required coverage level.

$$nTP = qAC * pACtp * \frac{1}{cobop}$$

```
\begin{equation}
nTP = qAC \cdot pAC_{tp} \cdot \frac{1}{cob_{op}}
\end{equation}
```

$$nTC = qAC * pACtc * \frac{cobop-1}{cobop}$$

```
\begin{equation}
nTC = qAC \cdot pAC_{tc} \cdot \frac{cob_{op} - 1}{cob_{op}}
\end{equation}
```

The ratio nTC/nTP is:

$$\frac{nTC}{nTP} = \frac{qAC * pACtc * \frac{cobop-1}{cobop}}{qAC * pACtp * \frac{1}{cobop}} = \frac{pACtc * (cobop-1)}{pACtp}$$

```
\begin{equation}
\frac{nTC}{nTP}
=
\frac{qAC \cdot pAC_{tc}\cdot \dfrac{cob_{op}-1}{cob_{op}}}
{qAC \cdot pAC_{tp}\cdot \dfrac{1}{cob_{op}}}
```

```
=
\frac{pAC_{tc}(cob_{op}-1)}{pAC_{tp}}
\end{equation}
```

$$nTC = \frac{pACtc*(cobop-1)}{pACtp} * nTP$$

```
\begin{equation}
nTC = \frac{pAC_{tc}(cob_{op}-1)}{pAC_{tp}} \cdot nTP
\end{equation}
```

$$nTP = \frac{pACtp}{pACtc*(cobop-1)} * nTC$$

```
\begin{equation}
nTP = \frac{pAC_{tp}}{pAC_{tc}(cob_{op}-1)} \cdot nTC
\end{equation}
```

A.8 Stabilization and Protected State Conditions

The protocol monitors the collateralization state of each Collateral Bag in order to determine whether the bucket is operating normally or has entered a protected state. When the collateralization state deteriorates beyond predefined limits, the protocol transitions the bucket into protected state and applies restrictions intended to preserve solvency.

Protected state threshold. Notation: prot_thrld

prot_thrld represents the protected state threshold parameter used by the protocol.

This parameter defines the coverage level below which a Collateral Bag enters protected state. When the collateralization of the bucket falls below prot_thrld, the protocol restricts operations that would further reduce the collateral backing the Token Peg liabilities assigned to that bucket.

Protected state conditions are evaluated using the collateral state variables of the Collateral Bag together with the coverage parameters defined earlier in this appendix. When the bucket is in protected state, the protocol may activate stabilization mechanisms that rebalance collateral and liabilities across buckets.

These stabilization mechanisms include the micro-liquidation and liquidation procedures described in the main body of the whitepaper.

A.9 Price Variables and Price Updates

Protocol operations that modify TP (USDRIF) liabilities or TC (RIFPRO,ROCR) balances rely on price variables that convert between TP (USDRIF) units and AC (RIF,DOC) units. These price variables appear in the atomic processes that update Peg Containers and Collateral Bags.

AC price in TPi. Notation: pACtpi

pACtpi represents the price of AC (RIF,DOC) expressed in TP (USDRIF) units for Peg Container i. This value is used during TP (USDRIF) issuance operations to determine how much collateral must be added to the Collateral Bag for a given amount of TP (USDRIF) issued.

AC price in TPi of the last operation. Notation: pACtp(i)lstop

pACtp(i)lstop represents the price of AC (RIF,DOC) expressed in TP (USDRIF) units recorded during the last operation affecting Peg Container i.

This variable is updated during the TP emission atomic process:

pACtplstop[i] = pACtpi

The stored value tracks the price used in the last operation affecting the Peg Container.

A.10 Oracle Inputs

Protocol operations depend on price inputs that represent the value of AC (RIF,DOC) used in the system. These values are provided to the protocol and are used in the conversion between TP (USDRIF) units and AC (RIF,DOC) units during issuance and collateral accounting updates.

AC price used in TP emission. Notation: pACtpi

pACtpi represents the price of AC (RIF,DOC) used when converting TP (USDRIF) issuance into collateral units in the TP emission atomic process.

TC price in AC. Notation: pTCacreal

pTCacreal represents the real-time price of TC (RIFPRO,ROCR) expressed in units of AC (RIF,DOC). This variable is used during TC (RIFPRO,ROCR) emission operations to determine how much collateral value is added to the Collateral Bag when new TC (RIFPRO,ROCR) is issued.

A.11 Accounting Relationships

The protocol maintains consistency between Peg Containers and Collateral Bags by applying atomic processes that update both TP (USDRIF) liabilities and collateral balances during issuance operations.

TP emission collateral update

nCAcb = nCAcb + (qTPi / pACtpi)

This relationship updates the collateral balance of the Collateral Bag by converting the issued TP (USDRIF) quantity into AC (RIF,DOC) units using the price variable pACtpi.

TC emission collateral update

nCAcb = nCAcb + (qTC / pTCacreal)

This relationship updates the collateral balance of the Collateral Bag during TC (RIFPRO,ROCR) issuance by converting the issued TC (RIFPRO,ROCR) quantity using the price variable pTCacreal.

A.12 Stabilization Mechanisms

The protocol includes stabilization procedures that rebalance collateral and liabilities across buckets when the collateralization state of a bucket deteriorates beyond acceptable limits. These procedures operate on the accounting variables defined in the previous sections and use the protected state threshold to determine when stabilization must occur.

Stabilization procedures adjust the distribution of TP (USDRIF) liabilities and AC (RIF,DOC) balances across Collateral Bags while preserving the total supply of TP (USDRIF).

Protected state threshold. Notation: prot_thrld

prot_thrld represents the threshold used by the protocol to determine when a Collateral Bag enters protected state.

When the collateralization state of a Collateral Bag falls below this threshold, the bucket is considered to be in protected state and stabilization procedures may be activated.

Collateral reallocation

Stabilization procedures may move collateral and liabilities between Collateral Bags in order to restore acceptable collateralization levels.

These operations modify the following accounting variables:

- nACcb — amount of AC (RIF,DOC) stored in the Collateral Bag
- nTP(i) — amount of TP (USDRIF) associated with Peg Container i
- nTCcb — amount of TC (RIFPRO,ROCR) issued by the Collateral Bag

The stabilization mechanisms described in the main body of the whitepaper operate on these variables in order to maintain system solvency.

A.13 Accounting Consistency Conditions

The protocol maintains internal consistency between Peg Containers and Collateral Bags through the accounting variables defined in this appendix. All protocol operations that modify TP (USDRIF) liabilities, TC (RIFPRO,ROCR) supply, or AC (RIF,DOC) balances must update these variables according to the atomic processes previously defined.

The following relationships must remain consistent throughout the protocol state.

Peg Container liability update

nTPi = nTPi + qTPi

This relationship updates the amount of TP (USDRIF) recorded in Peg Container i when new Token Peg is issued.

Collateral update from TP emission

nCAcb = nCAcb + (qTPi / pACtpi)

This relationship updates the collateral balance of the Collateral Bag by converting the issued TP (USDRIF) amount into AC (RIF,DOC) units using the price variable pACtpi.

Token Collateral issuance update

nTCcb = nTCcb + qTC

This relationship updates the amount of TC (RIFPRO,ROCR) issued by the Collateral Bag when new Token Collateral is minted.

Collateral update from TC emission

nCAcb = nCAcb + (qTC / pTCacreal)

This relationship updates the collateral balance of the Collateral Bag by converting the issued TC (RIFPRO,ROCR) amount using the price variable pTCacreal.

These accounting relationships ensure that Peg Container liabilities and Collateral Bag balances remain synchronized during protocol operations.

A.14 Collateral token price

Token Collateral represents the residual claim on the value of a bucket once the stablecoin liabilities attributed to that bucket have been satisfied. In balance-sheet terms, the collateral contained in the bucket acts as the asset side of the system, while the issued USDRIF represents its senior liability.

The value available to Token Collateral holders therefore corresponds to the remaining collateral value after accounting for the USDRIF liabilities assigned to the bucket. Dividing this residual value by the total supply of Token Collateral results in the implied price of the collateral token.

Formally, the Token Collateral price can be expressed as:

$$pTCac = \frac{nACcb - mocGain - \sum_{1}^{n} \frac{nTPi - tpGain}{pACtpi}}{nTCcb}$$

```
\begin{equation}
pTC_{ac} =
\frac{
nAC_{cb} - mocGain - \sum_{i=1}^{n} \frac{nTP_i - tpGain}{pACtp_i}
}
{nTC_{cb}}
\end{equation}
```

A.15 Collateral token leverage

Token Collateral behaves as a leveraged exposure to the underlying collateral asset. This arises from the residual claim structure of the bucket: stablecoin liabilities must be satisfied first, while Token Collateral holders absorb both gains and losses in the remaining value of the bucket.

As a result, proportional changes in the collateral asset value translate into amplified proportional changes in the value of Token Collateral.

This relationship can be expressed in terms of the bucket coverage ratio.

Let the coverage ratio be defined as:

$$C = \frac{nACcb \cdot P_{ca}}{nTP}$$

```
\begin{equation}
C = \frac{nACcb \cdot P_{AC}}{nTP}
\end{equation}
```

where the numerator represents the total value of collateral in the bucket and the denominator represents the value of Token Peg liabilities attributed to the bucket.

Under this balance-sheet structure, the effective leverage of Token Collateral relative to the collateral asset can be expressed as:

$$L_{ct} = \frac{C}{C-1}$$

```
\begin{equation}
 L_{ct} = \frac{C}{C - 1}
 \end{equation}
```

As the coverage ratio approaches one, the leverage of Token Collateral increases rapidly. Conversely, higher coverage ratios reduce leverage and provide a larger safety buffer for the system.

Appendix B — Flux Capacitor Formal Model

The Flux Capacitor is a transaction-dosing mechanism used to limit the rate and pattern of minting and redemption operations involving AC (RIF) in the RIF bucket.

Its purpose is to discourage and mitigate economic attacks that exploit the fact that minting TP (USDRIF) is economically similar to selling AC (RIF) without slippage, while redeeming TP (USDRIF) is economically similar to buying AC (RIF). Without restrictions, repeated large mint and redeem operations could create persistent arbitrage opportunities against external markets.

The Flux Capacitor restricts this behavior by applying limits to minting and redemption activity over a recent block window. These limits are computed using accumulators that track the recent transaction history and decay over time.

The Flux Capacitor operates independently for each bucket. In RIF On Chain, it is configured only for the RIF bucket.

B.1 Flux Capacitor Variables and Parameters

The Flux Capacitor uses the following variables and parameters.

Absolute Accumulator. Notation: AA

AA accumulates the absolute value of minting and redemption transactions. Both mint and redeem operations add their value to this accumulator.

Differential Accumulator. Notation: DA

DA accumulates the signed value of minting and redemption transactions. Mint operations add their value to the accumulator while redeem operations subtract their value.

Operational Difference. Notation: OD

OD represents the operational difference derived from the accumulators.

$OD = AA - |DA|$

This variable measures the amount of back-and-forth minting and redemption activity.

Block Number of the Last Accepted Transaction. Notation: BNLAT

BNLAT stores the block number of the last accepted transaction. It is used to determine how much decay must be applied to the accumulators.

Absolute Maximum Transaction Allowed. Notation: AMTA

AMTA represents the maximum absolute transaction volume allowed across the configured block span.

Maximum Operational Difference Allowed. Notation: MODA

MODA represents the maximum value allowed for the operational difference variable OD.

Block Span. Notation: BS

BS represents the number of blocks required for the decay factor to fully release transaction capacity.

B.2 Accumulator Mechanics

The Flux Capacitor tracks recent minting and redemption activity using two accumulators and a derived variable. These values are updated whenever a mint or redeem operation involving AC (RIF) is processed in the RIF bucket.

The accumulators measure two different aspects of recent activity: The total volume of mint and redeem operations, and the degree of back-and-forth activity between minting and redeeming

The Flux Capacitor enforces limits on both of these quantities using the parameters AMTA and MODA.

Absolute Accumulator update

The Absolute Accumulator (AA) tracks the total absolute volume of mint and redeem operations.

After a transaction with value MR (Mint or Redeem amount), the accumulator is updated as:

$AA = AA + |MR|$

This means both mint and redeem operations increase the accumulator by their absolute value.

The variable AA is compared against the parameter AMTA:

$|MR| \leq AMTA - AA$

If an operation would cause AA to exceed AMTA, the transaction is rejected.

This condition limits the total recent transaction volume that the protocol will accept.

Differential Accumulator update

The Differential Accumulator (DA) tracks the net direction of recent operations.

After a transaction with value MR:

DA = DA + MR

Mint operations increase DA, while redeem operations decrease it.

Unlike AA, the variable DA is not directly limited by a parameter. Instead it serves as an intermediate value used to compute the operational difference.

Operational Difference

The Operational Difference (OD) measures how much recent activity consists of alternating mint and redeem operations.

OD is defined as:

$OD = AA - |DA|$

This quantity becomes large when mint and redeem operations alternate frequently.

The value OD is compared against the parameter MODA:

$|OD| \leq MODA$

If an operation would cause the operational difference to exceed MODA, the transaction is rejected.

Interpretation of the Operational Difference

Because of the way OD is defined, the operational difference is effectively twice the value of the smaller of the two cumulative directions of activity.

In practice, this means:

$OD = 2 \times \min(\text{total mint volume}, \text{total redeem volume})$

As a result, the Flux Capacitor tolerates sustained activity in one direction (continuous minting or continuous redeeming) but limits repeated back-and-forth activity between minting and redemption.

This behavior discourages patterns of alternating mint and redeem operations that could otherwise be used to exploit short-term price discrepancies between the protocol and external markets.

B.3 Linear Decay

The Flux Capacitor gradually restores transaction capacity over time by applying a decay factor to the accumulator values. This mechanism ensures that limits enforced by the Flux Capacitor apply only to recent activity rather than permanently restricting the system.

The decay mechanism uses the number of blocks that have elapsed since the last accepted transaction to reduce the values of the accumulators before evaluating a new operation.

Elapsed blocks

Notation: n

n represents the number of blocks elapsed since the last accepted transaction.

$n = CB - BNLAT$

Where:

CB is the current block number

BNLAT is the Block Number of the Last Accepted Transaction

Linear Decay Factor

Notation: LDF

The Linear Decay Factor determines how much the accumulator values must be reduced before evaluating a new transaction.

$LDF = (-n / BS) + 1$

If the computed value becomes negative, it is clamped to zero:

if $LDF < 0 \rightarrow LDF = 0$

Where:

BS is the configured Block Span parameter

The Block Span represents the number of blocks required for the decay process to completely release previously consumed transaction capacity.

Decayed accumulator values

Before validating a new mint or redeem operation, the accumulators are updated using the decay factor.

$AA' = AA \times LDF$
$DA' = DA \times LDF$

These adjusted values represent the effective recent transaction activity after accounting for elapsed blocks.

If the candidate transaction is accepted, the protocol updates the block reference used by the decay mechanism:

BNLAT = CB

If the transaction is rejected, the decay adjustments are reverted and the stored accumulator values remain unchanged.

B.4 Transaction Acceptance Conditions

When a mint or redeem transaction involving AC (RIF) is proposed, the protocol evaluates whether the operation can be accepted under the Flux Capacitor limits. This evaluation occurs after applying the decay adjustments to the accumulators described in the previous section.

Let MR represent the value of the proposed mint or redeem operation.

Before evaluating the acceptance conditions, the accumulators are adjusted using the decay model:

$AA' = AA \times LDF$
$DA' = DA \times LDF$

The candidate transaction would produce the following updated accumulator values:

$AA'' = AA' + |MR|$
$DA'' = DA' + MR$

From these values, the resulting operational difference is computed:

$OD'' = AA' + |MR| - |DA' + MR|$

A transaction is accepted only if both of the following conditions are satisfied.

Absolute transaction limit

$|MR| \le AMTA - AA'$

This condition ensures that the total recent transaction volume does not exceed the maximum absolute transaction limit defined by AMTA.

Operational difference limit

$|OD''| \le MODA$

This condition ensures that the back-and-forth activity between minting and redeeming does not exceed the maximum operational difference allowed by MODA.

Acceptance rule

A transaction is accepted if and only if both conditions hold:

$(|MR| \le AMTA - AA')$ AND $(|OD''| \le MODA)$

If either condition fails, the transaction is rejected and the system state remains unchanged.

If the transaction is accepted, the accumulators are updated to the new values (AA'', DA'') and the block reference BNLAT is updated to the current block number.

B.5 Maximum Mint and Redeem Admitted (MMRA)

The acceptance conditions defined in the previous section determine the maximum value of a mint or redeem operation that can be accepted given the current accumulator state. This section derives the limits for mint and redeem operations based on those conditions.

Let MR represent the value of the proposed mint or redeem operation.

The acceptance conditions are:

① $|MR| \leq AMTA - AA'$

② $AA' + |MR| - |DA' + MR| \leq MODA$

These conditions determine the maximum amount that may be minted or redeemed depending on the current value of the decayed Differential Accumulator DA'.

Case 1: $DA' = 0$

When the differential accumulator is zero, the operational difference constraint does not impose any additional restriction beyond the absolute transaction limit.

$maxMint = AMTA - AA'$
$maxRedeem = AMTA - AA'$

Case 2: $DA' > 0$

In this case, recent activity has been predominantly minting.

$maxMint = AMTA - AA'$

$maxRedeem = \min(AMTA - AA' , (MODA - AA' + |DA'|) / 2)$

If the sign of DA' changes after applying the redeem operation, the operational difference constraint is replaced by the absolute transaction limit:

$maxRedeem = AMTA - AA'$

Case 3: $DA' < 0$

In this case, recent activity has been predominantly redeeming.

$maxRedeem = AMTA - AA'$

$\text{maxMint} = \min(\text{AMTA} - \text{AA}', (\text{MODA} - \text{AA}' + |\text{DA}'|) / 2)$

If the sign of DA′ changes after applying the mint operation, the operational difference constraint is replaced by the absolute transaction limit:

$\text{maxMint} = \text{AMTA} - \text{AA}'$

Non-negative constraint

If the computed value of maxMint or maxRedeem is negative, it is clamped to zero.

if $\text{maxMint} < 0 \rightarrow \text{maxMint} = 0$

if $\text{maxRedeem} < 0 \rightarrow \text{maxRedeem} = 0$

These expressions determine the maximum mint and redeem values that the protocol will accept for a transaction given the current accumulator state. The Flux Capacitor therefore limits both the total recent transaction volume and the degree of alternating mint and redeem activity.

B.6 Operational Notes and Implementation Considerations

The Flux Capacitor operates independently within each collateral bucket. In RIF On Chain, the mechanism is enabled only for the RIF bucket, where the underlying collateral asset is volatile. The DOC bucket does not apply Flux Capacitor limits.

Flux Capacitor validation occurs when queued operations are executed. Because minting and redemption requests are first placed in the Operation Queue, the Flux Capacitor is evaluated at execution time rather than when the request is initially submitted.

If an operation fails the Flux Capacitor conditions because its size alone exceeds the AMTA limit, the transaction is rejected. However, if the operation fails only because the current accumulator state temporarily exceeds the allowable limits, the transaction may remain in the queue and be executed later once the accumulators decay sufficiently.

Because decay continuously restores available capacity as blocks pass, queued operations that initially fail due to Flux Capacitor limits may later become executable without requiring user intervention.

Flux Capacitor parameters such as AMTA, MODA, and BS are ultimately controlled by protocol governance. In practice, operational control of these parameters has been revocably delegated to a multi-signature address operated by protocol automators, allowing adjustments to be made efficiently as market conditions evolve. Governance retains the authority to modify or revoke this delegation.

Changes to these parameters do not immediately re-evaluate operations already in the queue; queued transactions are evaluated only at the time of execution.

If a queued operation cannot yet be executed because Flux Capacitor limits are exceeded, the execution pipeline for that bucket pauses until the decay process restores sufficient capacity. Operations in other buckets remain unaffected.

In practice, the Flux Capacitor works together with other protections in the protocol — including the depth-weighted average price input, the operation queue, and mint and redemption fees — to discourage and mitigate attacks attempting to exploit price discrepancies between the protocol and external markets.

Appendix C — OMOC Oracle Infrastructure and RoC Jobs

C.1 Overview

The Money on Chain decentralized oracle infrastructure, referred to as OMOC, is a network of independent node operators that coordinate off-chain to perform jobs whose results are submitted to smart contracts on Rootstock.

An OMOC job may involve reaching consensus on external data, such as an asset price, or coordinating the execution of a predefined on-chain function. In both cases, the validity and consequences of the submitted transaction are determined by smart-contract rules.

OMOC operators pay the RBTC gas required to submit their assigned transactions and receive rewards denominated in MOC tokens.

C.2 Operator Participation and Staking

Running an OMOC node requires the operator to stake MOC tokens. The stake provides economic commitment to the oracle system and determines which registered operators may actively participate.

Only the ten eligible nodes with the largest active stakes participate in consensus and receive rewards during a round. Operators outside the active set may increase their stake and become eligible in a subsequent round if they enter the ten highest-ranked positions.

This model creates competition for participation while requiring active operators to maintain meaningful economic exposure to the system they support.

C.3 Rounds and Publisher Rotation

OMOC operates in rounds lasting approximately 30 days.

During each round, participating nodes rotate responsibility for performing the jobs to which they are subscribed. These responsibilities include publishing agreed price data and executing periodic protocol tasks. Rotation distributes operational responsibility among the active nodes and prevents the system from depending permanently on a single publisher.

At the end of a round, eligibility and participation may be recalculated according to the applicable staking and subscription rules.

C.4 Jobs Relevant to RIF On Chain

RoC relies on three OMOC jobs.

RIF/USD

The RIF/USD job publishes the market price of RIF expressed in U.S. dollars.

RoC uses this price directly to:

- value the collateral held by the RIF bucket;
- calculate the bucket's coverage ratio;
- evaluate issuance and redemption constraints;
- calculate EMA-adjusted coverage parameters where applicable; and
- determine whether stabilization conditions have been reached.

BTC/USD

The BTC/USD job publishes the market price of Bitcoin expressed in U.S. dollars.

RoC does not ordinarily use this price directly in its own bucket accounting. It depends on it indirectly because DOC is issued by the Money on Chain protocol and is backed by Bitcoin collateral. The BTC/USD price is required by Money on Chain to evaluate its collateralization and preserve the conditions supporting DOC's USD peg.

Because DOC serves as collateral in the RoC DOC bucket, the reliable operation of the BTC/USD job forms part of the infrastructure on which RoC's DOC-backed issuance depends.

TasksRunner

TasksRunner participants invoke protocol functions that must be executed periodically or after their on-chain conditions have been satisfied. These functions include:

- processing queued mint and redemption operations;
- performing required periodic accounting operations;
- triggering micro-liquidation when its contract conditions are met;
- triggering liquidation when its contract conditions are met; and
- executing other maintenance functions assigned to the job by governance.

TasksRunner operators do not determine whether an operation is valid or choose its economic result. Those decisions are encoded in the RoC smart contracts. The operator submits the transaction, while the contracts verify the applicable conditions and perform the corresponding state transition.

C.5 User-Funded Execution

Mint and redemption requests are submitted through an operations queue and executed later by TasksRunner operators.

When a user enqueues an operation, the user is charged an additional amount of RBTC intended to cover the on-chain execution costs associated with that request. This amount funds:

- the gas required to execute the queued operation;

- the provision of a sufficiently recent RIF/USD price when required; and
- the provision of a sufficiently recent BTC/USD price when required.

The collected execution amount is separate from the protocol's percentage-based minting or redemption fee. Its purpose is to compensate for the Rootstock transaction costs incurred in making the requested operation executable.

The precise amount may depend on estimated gas requirements, current network gas prices, and which oracle publications are required before execution.

C.6 Protocol-Funded Rewards

In addition to user-funded execution payments, OMOC jobs receive defined portions of RoC protocol revenue.

RIF/USD providers

Operators providing the RIF/USD price receive:

- 15% of minting and redemption fees generated by the RIF bucket; and
- 15% of minting and redemption fees generated by the DOC bucket.
- 15% of the periodic interest collected from RIFPRO holders.

TasksRunner executors

Operators executing the TasksRunner job receive:

- 0.75% of minting and redemption fees generated by the RIF bucket;
- 0.75% of minting and redemption fees generated by the DOC bucket; and
- 1.5% of the periodic interest collected from RIFPRO holders.

BTC/USD providers

Operators providing the BTC/USD price receive:

- 6.75% of minting and redemption fees generated by the RIF bucket;
- 6.75% of minting and redemption fees generated by the DOC bucket; and
- 13.5% of the periodic interest collected from RIFPRO holders.

These allocations may be modified through the applicable governance process.

C.7 Economic Security and Operational Continuity

RoC depends on timely price publication and task execution. Its incentive system must therefore make sustained OMOC participation economically attractive relative to operators' infrastructure costs, gas expenditure, capital requirements, and operational responsibilities.

The staking requirement creates economic commitment, while job rewards compensate operators for providing infrastructure and submitting transactions. The selection of an active operator set and the rotation of publishing responsibilities reduce dependence on any single node.

A healthy decentralized governance process must periodically evaluate whether these incentives remain sufficient to attract qualified operators, preserve competition for active positions, and maintain reliable protocol operation without interruption.

Additional general information about OMOC is available through the Money on Chain oracle documentation at https://moneyonchain.com/omoc-oracles/